

HIRAMOTO Toshiro

professor

基本信息

姓名: HIRAMOTO Toshiro

所属院校: 东京大学

地区: Japan

学科领域

微电子科学与工程

集成电路设计与集成系统

电子科学与技术

研究领域

微电子器件与制造

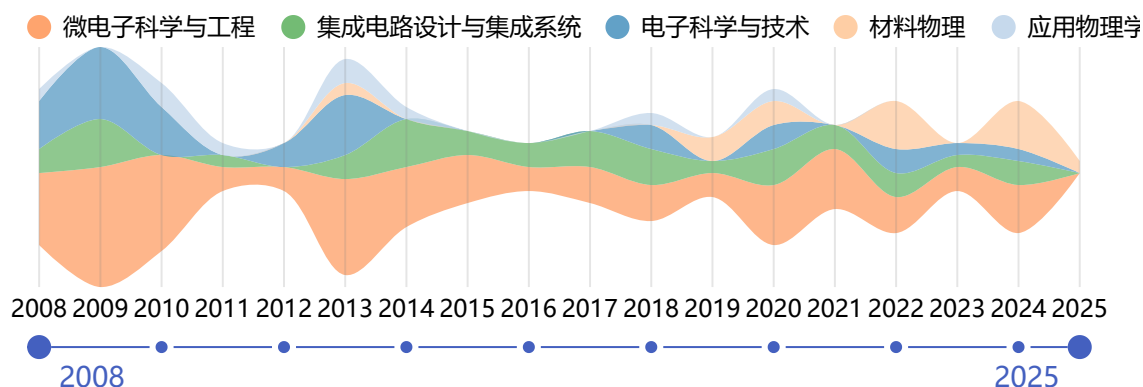
纳米电子与先进工艺

半导体器件物理与工艺

个人简介

Professor HIRAMOTO Toshiro specializes in Integrated Device Engineering within the Department of Informatics and Electronics. His research at the Electrical Engineering and Information Systems Graduate School of Engineering focuses on Very Large Scale Integration (VLSI) and future integrated nanoelectronics. The Hiramoto/Kobayashi Lab aims to address global issues through technological innovation, particularly in low voltage and low power CMOS and VLSI, SOI, FinFET, Body effect, and Double-gate FET.

近年发表的论文



Enhanced reliability of ferroelectric Hf_{0.5}Zr_{0.5}O₂ capacitors by bottom electrode surface oxidation

JAPANESE JOURNAL OF APPLIED PHYSICS

Itoya, Yuki; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 0

2025

Robust reverse bias safe operating area and improved electrical performance in 3300 V non-proportionally scaled insulated gate bipolar transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Zhou, Xiang; Fukui, Munetoshi; Takeuchi, Kiyoshi; Saraya, Takuya; Saito, Wataru; Hiramoto, Toshiro

被引用次数: 0

2024

Oxide-semiconductor channel ferroelectric field-effect transistors for high-density memory applications: 3D NAND operation and the potential impact of in-plane polarization

JAPANESE JOURNAL OF APPLIED PHYSICS

Hao, Junxiang; Mei, Xiaoran; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 0

2024

High-Field Transport and Statistical Variability of Nanosheet Oxide Semiconductor FETs With Channel Length Scaling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Huang, Xingyu; Hikake, Kaito; Kim, Sung-Hun; Sakai, Kota; Li, Zhuo; Mizutani, Tomoko; Saraya, Takuya; Hiramoto, Toshiro; Takahashi, Takanori; Uenuma, Mutsunori; Uraoka, Yukiharu; Kobayashi, Masaharu

被引用次数: 0

2024

A Nanosheet Oxide Semiconductor FET Using ALD InGaO_x Channel for 3-D Integrated Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hikake, Kaito; Li, Zhuo; Hao, Junxiang; Pandey, Chitra; Saraya, Takuya; Hiramoto, Toshiro; Takahashi, Takanori; Uenuma, Mutsunori; Uraoka, Yukiharu; Kobayashi, Masaharu

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2024

Superior Turn-Off dV/dt Controllability From Suppression of Dynamic Avalanche in 3300V Scaled IGBTs

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Zhou, Xiang; Fukui, Munetoshi; Takeuchi, Kiyoshi; Saraya, Takuya; Hiramoto, Toshiro

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2024

Energy-Efficient Annealing Process of Ferroelectric Hf_{0.5}Zr_{0.5}O₂ Capacitor Using Ultraviolet-LED for Green Manufacturing

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Yamada, Hirotaka; Furue, Satoru; Yokomori, Takehiko; Itoya, Yuki; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 0

2024

Future Perspectives of CMOS Logic Innovation Beyond 2nm

2024 IEEE SILICON NANOELECTRONICS WORKSHOP, SNW 2024

Hiramoto, Toshiro; Wakabayashi, Hitoshi

被引用次数: 0

2024

A Nanosheet Oxide Semiconductor FET Using ALD InZnOx Channel

2024 IEEE SILICON NANOELECTRONICS WORKSHOP, SNW 2024

Kim, Sung-hun; Hikake, Kaito; Li, Zhuo; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

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2024

Reducing turn-off loss by suppressing back-side parasitic npn transistor in single-back and double-front Gate-controlled IGBT

2024 36TH INTERNATIONAL SYMPOSIUM ON POWER SEMICONDUCTOR DEVICES AND IC S, ISPSD 2024

Kobayashi, Yusuke; Fukui, Munetoshi; Matsudai, Tomoko; Saraya, Takuya; Itou, Kazuo; Takakura, Toshihiko; Suzuki, Shinichi; Sakano, Tatsunori; Yamamoto, Takato; Inokuchi, Tomoaki; Gejo, Ryohei; Kobayashi, Kenya; Takao, Kazuto; Hiramoto, Toshiro

被引用次数: 0

2024

Cryogenic threshold voltage and on-current variability comparative analysis of same-fab 65 nm bulk and fully depleted silicon-on-insulator metal-oxide-semiconductor field-effect transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Liu, Zihao; Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Oka, Hiroshi; Mori, Takahiro; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2024

Performance and reliability of nanosheet oxide semiconductor FETs with ALD-grown InGaO for 3D integration (Invited)

2024 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM, IRPS 2024

Kobayashi, Masaharu; Hikake, Kaito; Li, Zhuo; Hao, Junxiang; Pandey, Chitra; Saraya, Takuya; Hiramoto, Toshiro; Takahashi, Takanori; Uenuma, Mutsunori; Uraoka, Yukiharu

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2024

Device modeling of oxide-semiconductor channel antiferroelectric FETs using half-loop hysteresis for memory operation

JAPANESE JOURNAL OF APPLIED PHYSICS

Huang, Xingyu; Itoya, Yuki; Li, Zhuo; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

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2023

MOSFET series resistance extraction at cryogenic temperatures

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Mizutani, Tomoko; Saraya, Takuya; Oka, Hiroshi; Mori, Takahiro; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 1

2023

On the thickness dependence of the polarization switching kinetics in HfO₂-based ferroelectric

APPLIED PHYSICS LETTERS

Sawabe, Yoshiki; Saraya, Takuya; Hiramoto, Toshiro; Su, Chun-Jung; Hu, Vita Pi-Ho; Kobayashi, Masaharu

被引用次数: 19

2022

Estimation of minimum operating voltage in fully depleted SOI SRAM cells using gamma distribution

JAPANESE JOURNAL OF APPLIED PHYSICS

Yu, Hongkuan; Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2022

A simulation study on memory characteristics of InGaZnO-channel ferroelectric FETs with 2D planar and 3D structures

JAPANESE JOURNAL OF APPLIED PHYSICS

Mo, Fei; Mei, Xiaoran; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 15

2022

A robust single device MOSFET series resistance extraction method considering horizontal-field-dependent mobility

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 4

2022

Effect of percolation path on temperature dependence of threshold voltage variability in bulk MOSFETs

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Oka, Hiroshi; Mori, Takahiro; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 4

2022

Cause analysis of width-dependence of on-current variability in thin gate-all-around silicon nanowire MOSFET

JAPANESE JOURNAL OF APPLIED PHYSICS

Liu, Zihao; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2022

A robust single device MOSFET series resistance extraction method considering horizontal-field-dependent mobility (vol 61, SC1016, 2022)

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2022

A 3D Vertical-Channel Ferroelectric/Anti-Ferroelectric FET With Indium Oxide

IEEE ELECTRON DEVICE LETTERS

Li, Zhuo; Wu, Jixuan; Mei, Xiaoran; Huang, Xingyu; Saraya, Takuya; Hiramoto, Toshiro; Takahashi, Takanori; Uenuma, Mutsunori; Uraoka, Yukiharu; Kobayashi, Masaharu

被引用次数: 26

2022

Ultrathin MoS₂-Channel FeFET Memory With Enhanced Ferroelectricity in HfZrO₂ and Body-Potential Control

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Xiang, Jiawen; Chang, Wen Hsin; Saraya, Takuya; Hiramoto, Toshiro; Irisawa, Toshifumi; Kobayashi, Masaharu

被引用次数: 7

2022

Efficient Erase Operation by GIDL Current for 3D Structure FeFETs With Gate Stack Engineering and Compact Long-Term Retention Model

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Mo, Fei; Xiang, Jiawen; Mei, Xiaoran; Sawabe, Yoshiki; Saraya, Takuya; Hiramoto, Toshiro; Su, Chun-Jung; Hu, Vita Pi-Ho; Kobayashi, Masaharu

被引用次数: 5

2022

Variability characteristics and corner effects of gate-all-around (GAA) p-type poly-Si junctionless nanowire/nanosheet transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Ahn, Min-Ju; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2021

Design space exploration of hysteretic negative capacitance ferroelectric FETs based on static solutions of Landau-Khalatnikov model for nonvolatile memory applications

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2021

Reconfigurable Multivalued Logic Functions of a Silicon Ellipsoidal Quantum-Dot Transistor Operating at Room Temperature

ACS NANO

Lee, Youngmin; Lee, Jin Woo; Lee, Sejoon; Hiramoto, Toshiro; Wang, Kang L.

被引用次数: 10

2021

Subthreshold Swing in Silicon Gate-All-Around Nanowire and Fully Depleted SOI MOSFETs at Cryogenic Temperature

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Sekiguchi, Shohei; Ahn, Min-Ju; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 7

2021

Analysis of Drain Current Variability Components in Extremely Narrow GAA Silicon Nanowire MOSFETs of 4nm Width

2021 SILICON NANO-ELECTRONICS WORKSHOP (SNW)

Liu, Zihao; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2021

A first-principles study on ferroelectric phase formation of Si-doped HfO₂ through nucleation and phase transition in thermal process

APPLIED PHYSICS LETTERS

Wu, Jixuan; Mo, Fei; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 34

2020

Origin of carrier lifetime degradation in floating-zone silicon during a high-temperature process for insulated gate bipolar transistor

JAPANESE JOURNAL OF APPLIED PHYSICS

Yokogawa, Ryo; Kobayashi, Hiroto; Numasawa, Yohichiroh; Ogura, Atsushi; Nishizawa, Shin-ichi; Saraya, Takuya; Ito, Kazuo; Takakura, Toshihiko; Suzuki, Shinichi; Fukui, Munetoshi; Takeuchi, Kiyoshi; Hiramoto, Toshiro

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2020

Superior subthreshold characteristics of gate-all-around p-type junctionless poly-Si nanowire transistor with ideal subthreshold slope

JAPANESE JOURNAL OF APPLIED PHYSICS

Ahn, Min-Ju; Saraya, Takuya; Kobayashi, Masaharu; Sawamoto, Naomi; Ogura, Atsushi; Hiramoto, Toshiro

被引用次数: 9

2020

Statistical analysis of temperature dependence of worst case static random access memory data retention voltage using extreme value theory

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2020

Impact of structural parameter scaling on on-state voltage in 1200 V scaled IGBTs

JAPANESE JOURNAL OF APPLIED PHYSICS

Saraya, Takuya; Itou, Kazuo; Takakura, Toshihiko; Fukui, Munetoshi; Suzuki, Shinichi; Takeuchi, Kiyoshi; Kakushima, Kuniyuki; Hoshii, Takuya; Tsutsui, Kazuo; Iwai, Hiroshi; Nishizawa, Shin-ichi; Omura, Ichiro; Hiramoto, Toshiro

被引用次数: 3

2020

A simulation study on low voltage operability of hafnium oxide based ferroelectric FET memories

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 5

2020

Width dependence of drain current and carrier mobility in gate-all-around multi-channel polycrystalline silicon nanowire transistors with 10nm width scale

JAPANESE JOURNAL OF APPLIED PHYSICS

Jang, Ki-Hyun; Saraya, Takuya; Kobayashi, Masaharu; Sawamoto, Naomi; Ogura, Atsushi; Hiramoto, Toshiro

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2020

Low-Voltage Operating Ferroelectric FET with Ultrathin IGZO Channel for High-Density Memory Application

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Mo, Fei; Tagawa, Yusaku; Jin, Chengji; Ahn, MinJu; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 86

2020

Physical Mechanisms of Reverse DIBL and NDR in FeFETs With Steep Subthreshold Swing

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Jin, Chengji; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 19

2020

Bipolar Transistor Test Structures for Extracting Minority Carrier Lifetime in IGBTs

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Takeuchi, Kiyoshi; Fukui, Munetoshi; Saraya, Takuya; Itou, Kazuo; Takakura, Toshihiko; Suzuki, Shinichi; Numasawa, Yohichiroh; Shigyo, Naoyuki; Kakushima, Kuniyuki; Hoshii, Takuya; Furukawa, Kazuyoshi; Watanabe, Masahiro; Wakabayashi, Hitoshi; Tsutsui, Kazuo; Iwai, Hiroshi; Ogura, Atsushi; Saito, Wataru; Nishizawa, Shin-Ichi; Tsukuda, Masanori; Omura, Ichiro; Ohashi, Hiromichi; Hiramoto, Toshiro

被引用次数: 1

2020

Reduced variability of drain-induced barrier lowering and subthreshold slope at high temperature in bulk and silicon-on-thin-buried-oxide (SOTB) MOSFETs

JAPANESE JOURNAL OF APPLIED PHYSICS

Gao, Shuang; Mizutani, Tomoko; Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 4

2019

Evaluations of minority carrier lifetime in floating zone Si affected by Si insulated gate bipolar transistor processes

JAPANESE JOURNAL OF APPLIED PHYSICS

Kobayashi, Hiroto; Yokogawa, Ryo; Kinoshita, Kosuke; Numasawa, Yohichiroh; Ogura, Atsushi; Nishizawa, Shin-ichi; Saraya, Takuya; Ito, Kazuo; Takakura, Toshihiko; Suzuki, Shin-ichi; Fukui, Munetoshi; Takeuchi, Kiyoshi; Hiramoto, Toshiro

被引用次数: 2

2019

Ferroelectric HfO₂ Tunnel Junction Memory With High TER and Multi-Level Operation Featuring Metal Replacement Process

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Kobayashi, Masaharu; Tagawa, Yusaku; Mo, Fei; Saraya, Takuya; Hiramoto, Toshiro

被引用次数: 94

2019

Triple-Stacked Au/SiO₂ Hybrid Bonding With 6- μ m-Pitch Au Electrodes on Silicon-on-Insulator Substrates Using O₂ Plasma Surface Activation for 3-D Integration

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Honda, Yuki; Goto, Masahide; Watabe, Toshihisa; Hagiwara, Kei; Nanba, Masakazu; Iguchi, Yoshinori; Saraya, Takuya; Kobayashi, Masaharu; Higurashi, Eiji; Toshiyoshi, Hiroshi; Hiramoto, Toshiro

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2019

Role of gate current and polarization switching in sub-60 mV/decade steep subthreshold slope in metal-ferroelectric HfZrO₂-metal-insulator-Si FET

JAPANESE JOURNAL OF APPLIED PHYSICS

Jang, Kyungmin; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 3

2018

Lowering data retention voltage in static random access memory array by post fabrication self-improvement of cell stability by multiple stress application

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2018

Optimizing MOS-gated thyristor using voltage-based equivalent circuit model for designing steep-subthreshold-slope PN-body-tied silicon-on-insulator FET

JAPANESE JOURNAL OF APPLIED PHYSICS

Ueda, Daiki; Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

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2018

On gate stack scalability of double-gate negative-capacitance FET with ferroelectric HfO₂ for energy efficient sub-0.2V operation

JAPANESE JOURNAL OF APPLIED PHYSICS

Jang, Kyungmin; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2018

Experimental Demonstration of a Nonvolatile SRAM With Ferroelectric HfO₂ Capacitor for Normally Off Application

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Kobayashi, Masaharu; Ueyama, Nozomu; Jang, Kyungmin; Hiramoto, Toshiro

被引用次数: 11

2018

Experimental Observation and Simulation Model for Transient Characteristics of Negative-Capacitance in Ferroelectric HfZrO₂ Capacitor

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Jang, Kyungmin; Ueyama, Nozomu; Kobayashi, Masaharu; Hiramoto, Toshiro

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2018

Ion/loff ratio enhancement and scalability of gate-all-around nanowire negative-capacitance FET with ferroelectric HfO₂

SOLID-STATE ELECTRONICS

Jang, Kyungmin; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 29

2017

Correlation between static random access memory power-up state and transistor variation

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Mizutani, Tomoko; Saraya, Takuya; Shinohara, Hirofumi; Kobayashi, Masaharu; Hiramoto, Toshiro

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2017

Parallel programmable nonvolatile memory using ordinary static random access memory cells

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Shinohara, Hirofumi; Kobayashi, Masaharu; Hiramoto, Toshiro

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2017

Measurement of Static Random Access Memory Power-Up State Using an Addressable Cell Array Test Structure

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Takeuchi, Kiyoshi; Mizutani, Tomoko; Shinohara, Hirofumi; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2017

Negative Capacitance for Boosting Tunnel FET performance

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Kobayashi, Masaharu; Jang, Kyungmin; Ueyama, Nozomu; Hiramoto, Toshiro

被引用次数: 88

2017

On device design for steep-slope negative-capacitance field-effect-transistor operating at sub-0.2V supply voltage with ferroelectric HfO₂ thin film

AIP ADVANCES

Kobayashi, Masaharu; Hiramoto, Toshiro

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2016

Transistor-level characterization of static random access memory bit failures induced by random telegraph noise

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Saraya, Takuya; Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

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2016

Multiple logic functions from extended blockade region in a silicon quantum-dot transistor

JOURNAL OF APPLIED PHYSICS

Lee, Youngmin; Lee, Sejoon; Im, Hyunsik; Hiramoto, Toshiro

被引用次数: 9

2015

Detailed analysis of minimum operation voltage of extraordinarily unstable cells in fully depleted silicon-on-buried-oxide six-transistor static random access memory

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Yamamoto, Yoshiki; Makiyama, Hideki; Yamashita, Tomohiro; Oda, Hidekazu; Kamohara, Shiro; Sugii, Nobuyuki; Hiramoto, Toshiro

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2015

Comparison and statistical analysis of four write stability metrics in bulk CMOS static random access memory cells

JAPANESE JOURNAL OF APPLIED PHYSICS

Qiu, Hao; Mizutani, Tomoko; Saraya, Takuya; Hiramoto, Toshiro

被引用次数: 17

2015

Nanoelectronics Research Gaps and Recommendations A Report from the International Planning Working Group on Nanoelectronics (IPWGN)

IEEE TECHNOLOGY AND SOCIETY MAGAZINE

Galatsis, Kosmas; Gargini, Paolo; Hiramoto, Toshiro; Beernaert, Dirk; DeKeersmaecker, Roger; Pelka, Joachim; Pfitzner, Lothar

被引用次数: 5

2015

Effect of drain-induced barrier lowering on performance of ultralow-supply-voltage CMOS circuits operating in subthreshold region

JAPANESE JOURNAL OF APPLIED PHYSICS

Jung, Seung-Min; Mizutani, Tomoko; Hiramoto, Toshiro

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2014

Threshold voltage shifts and their variability behaviors in p-channel FETs by high voltage on-state and off-state stress

JAPANESE JOURNAL OF APPLIED PHYSICS

Alias, Nurul Ezaila; Kumar, Anil; Saraya, Takuya; Hiramoto, Toshiro

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2014

Comparison and distribution of minimum operation voltage in fully depleted silicon-on-thin-buried-oxide and bulk static random access memory cells

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Yamamoto, Yoshiki; Makiyama, Hideki; Shinohara, Hirofumi; Iwamatsu, Toshiaki; Oda, Hidekazu; Sugii, Nobuyuki; Hiramoto, Toshiro

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2014

Peak position control of Coulomb blockade oscillations in silicon single-electron transistors with floating gate operating at room temperature

JAPANESE JOURNAL OF APPLIED PHYSICS

Tanahashi, Yuma; Suzuki, Ryota; Saraya, Takuya; Hiramoto, Toshiro

被引用次数: 3

2014

Observation of Single Electron Transport via Multiple Quantum States of a Silicon Quantum Dot at Room Temperature

NANO LETTERS

Lee, Sejoon; Lee, Youngmin; Song, Emil B.; Hiramoto, Toshiro

被引用次数: 59

2014

A novel MOSFET with vertical signal-transfer capability for 3D-structured CMOS image sensors

IEEJ TRANSACTIONS ON ELECTRICAL AND ELECTRONIC ENGINEERING

Goto, Masahide; Hagiwara, Kei; Iguchi, Yoshinori; Ohtake, Hiroshi; Saraya, Takuya; Toshiyoshi, Hiroshi; Hiramoto, Toshiro

被引用次数: 8

2014

Modulation of peak-to-valley current ratio of Coulomb blockade oscillations in Si single hole transistors

APPLIED PHYSICS LETTERS

Lee, Sejoon; Lee, Youngmin; Song, Emil B.; Hiramoto, Toshiro

被引用次数: 12

2013

Nonvolatile three-terminal operation based on oxygen vacancy drift in a Pt/Ta₂O_{5-x}/Pt, Pt structure

APPLIED PHYSICS LETTERS

Wang, Qi; Itoh, Yaomi; Hasegawa, Tsuyoshi; Tsuruoka, Tohru; Yamaguchi, Shu; Watanabe, Satoshi; Hiramoto, Toshiro; Aono, Masakazu

被引用次数: 18

2013

Experimental Observation of Quantum Confinement Effect in (110) and (100) Silicon Nanowire Field-Effect Transistors and Single-Electron/Hole Transistors Operating at Room Temperature

JAPANESE JOURNAL OF APPLIED PHYSICS

Suzuki, Ryota; Nozue, Motoki; Saraya, Takuya; Hiramoto, Toshiro

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2013

Effects of Side Surface Roughness on Carrier Mobility in Tri-Gate Single Silicon Nanowire Metal-Oxide-Semiconductor Field-Effect Transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Mao, Ke; Saraya, Takuya; Hiramoto, Toshiro

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2013

Direct Measurement of Carrier Mobility in Intrinsic Channel Tri-Gate Single Silicon Nanowire Metal-Oxide-Semiconductor Field-Effect Transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Mao, Ke; Saraya, Takuya; Hiramoto, Toshiro

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2013

Statistical Analysis of Subthreshold Swing in Fully Depleted Silicon-on-Thin-Buried-Oxide and Bulk Metal-Oxide-Semiconductor Field Effect Transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Yamamoto, Yoshiki; Makiyama, Hideki; Iwamatsu, Toshiaki; Oda, Hidekazu; Sugii, Nobuyuki; Hiramoto, Toshiro

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2013

Integration of Complementary Metal-Oxide-Semiconductor 1-Bit Analog Selectors and Single-Electron Transistors Operating at Room Temperature

JAPANESE JOURNAL OF APPLIED PHYSICS

Suzuki, Ryota; Nozue, Motoki; Saraya, Takuya; Hiramoto, Toshiro

被引用次数: 4

2013

Experimental Demonstration of Post-Fabrication Self-Improvement of SRAM Cell Stability by High-Voltage Stress

IEICE TRANSACTIONS ON ELECTRONICS

Hiramoto, Toshiro; Kumar, Anil; Saraya, Takuya; Miyano, Shinji

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2013

NBTI Reliability of PFETs under Post-Fabrication Self-Improvement Scheme for SRAM

IEICE TRANSACTIONS ON ELECTRONICS

Alias, Nurul Ezaila; Kumar, Anil; Saraya, Takuya; Miyano, Shinji; Hiramoto, Toshiro

被引用次数: 4

2013

Statistical Analysis of Current Onset Voltage (CM) Distribution of Scaled MOSFETs

IEICE TRANSACTIONS ON ELECTRONICS

Mizutani, Tomoko; Kumar, Anil; Hiramoto, Toshiro

被引用次数: 3

2013

Gate Length and Gate Width Dependence of Drain Induced Barrier Lowering and Current-Onset Voltage Variability in Bulk and Fully Depleted Silicon-on-Insulator Metal Oxide Semiconductor Field Effect Transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Kumar, Anil; Mizutani, Tomoko; Hiramoto, Toshiro

被引用次数: 7

2012

Suppression of Within-Device Variability in Intrinsic Channel Tri-Gate Silicon Nanowire Metal-Oxide-Semiconductor Field-Effect Transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Mao, Ke; Mizutani, Tomoko; Kumar, Anil; Saraya, Takuya; Hiramoto, Toshiro

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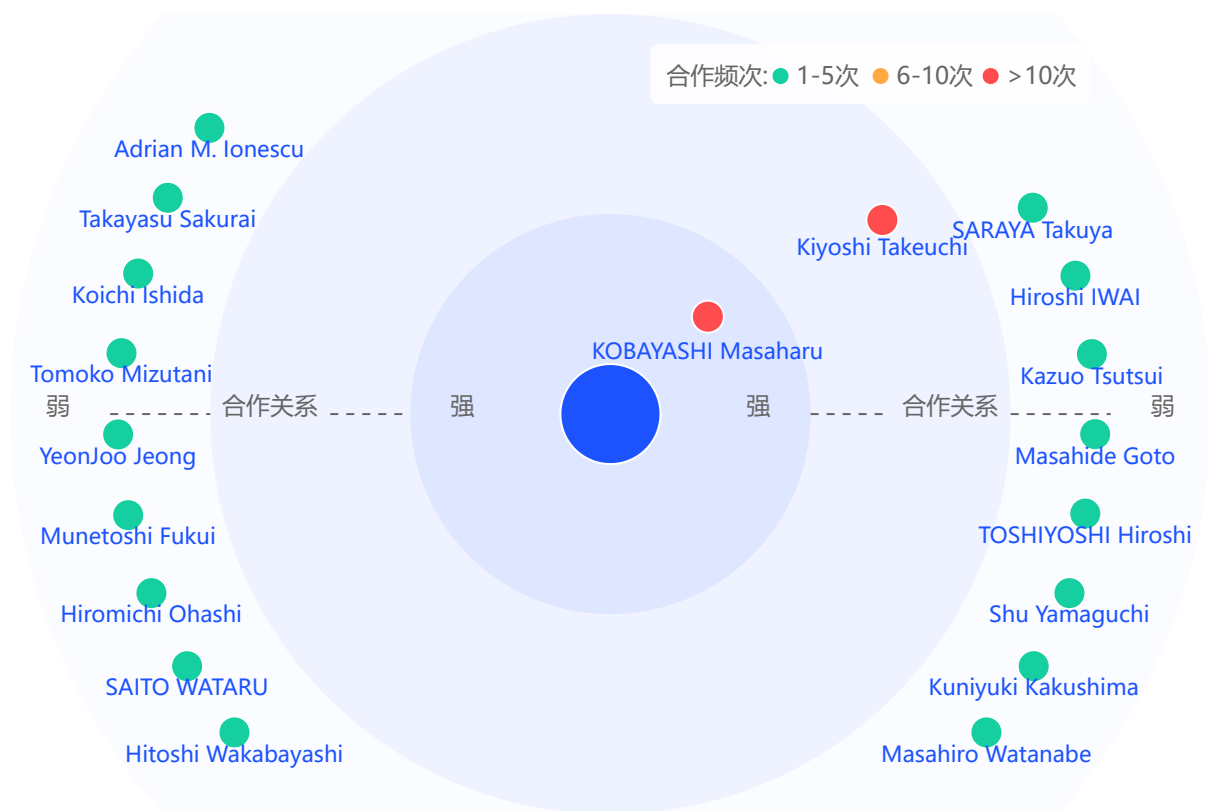
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